



Quasi-Solid Dye Sensitized Solar Cell with Straight Ion Paths Proposal of Hybrid Electrolytes for Ionic Liquid-Type Electrolytes

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Quasi-solid dye sensitized solar cells (QDSC) with straight ion paths are reported. The cell consists of anodic oxidation Al_2O_3 films. Ionic liquid-type electrolyte is filled in the straight nanopores and they are sandwiched between a counter electrode and a TiO_2 electrode. Even when electrochemically inactive Al_2O_3 wall occupies 50% of the electrolyte layer, the I_3^- diffusion coefficient and solar cell performance are the same as those without the Al_2O_3 layers. This contrasts the results of cells filled with the mixture of Al_2O_3 nanoparticles and ionic liquid type electrolytes, where random ion paths are expected. In addition, when the inner wall of anodic oxidation Al_2O_3 films is chemically modified with long alkyl groups bearing methylimidazolium moieties, the increase in photovoltaic performance is observed. This is explained by the existence of ion paths along aligned methylimidazolium moieties in the surface-modified Al_2O_3 films, where $\text{I}_2(\text{I}_3^-)$ species are concentrated. The swift apparent I_3^- diffusion coefficient is explained by Gröthuss mechanism. Because the I_2 species are concentrated in the straight pore of Al_2O_3 layers, the concentration of $\text{I}_2(\text{I}_3^-)$ species in TiO_2 layers is reduced relatively. This increases light penetration depth at around 400 nm in the porous TiO_2 layer and improves incident photon-to-current efficiency at around 400 nm. The hybrid electrolyte systems containing both of concentrated $\text{I}_2(\text{I}_3^-)$ in a porous Al_2O_3 layer and low concentration portions in nanopores in TiO_2 are proposed to increase ionic liquid-type QDSC.

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Dye sensitized solar cells (DSC) are expected to be one of the promising solar cells in the near future because of the high photo-energy conversion efficiency ($>10\%$) and the easy cell preparation processes.¹ One of the essential research items is solidification. All-solid state DSCs²⁻¹² and quasi-solid DSCs filled with gel electrolytes¹³⁻²⁶ have been reported. Instead of volatile organic-type gel electrolytes, gel electrolytes consisting of less volatile ionic liquids have been reported.²⁷⁻³³ Recently, composite gels consisting of nanoparticles and ionic liquids have been reported because of the easy preparation and less decrease in solar cell performances after solidification.^{24-26,34} However, the nanoparticle content was restricted to less than 10%, because the increase in the content of the electrochemically inactive nanoparticles retards swift ion diffusion. Therefore, these gel electrolytes consisting of nanoparticles were soft gel. We have already reported hard clay type gel electrolytes, where, nanoparticles were surface-modified with imidazolium moieties bearing a long alkyl group.³⁵ Because of the ion paths fabricated along the imidazolium moieties bonded to TiO_2 nanoparticles through a long alkyl group, the apparent ion-diffusion coefficient and photovoltaic performances after solidification were the same as that before the solidification even when the content of the nanoparticles reached 100% against the ionic liquid-type electrodes (hard clay type gel). We made clear that the long alkyl group is necessary to align the imidazolium moieties by self-organization of the alkyl groups, a kind of liquid crystalline properties. In this model, ion diffusion was expected to occur randomly (Fig. 1A and Fig. 2 cell 2). To realize straight one-directional diffusion from a cathode to an anode or from an anode to a cathode, we use anodic oxidation Al_2O_3 films, where nanopores penetrate the film straightly from one side to the other side (Fig. 1B). The film was inserted between a TiO_2 anode and a counter cathode and the pore was filled with ionic liquid-type electrolytes (Fig. 2, cell 3). In this report, the photovoltaic performance for the cell consisting of porous Al_2O_3 films are reported. The increase in the photovoltaic performance is explained by straight ion diffusion and Gröthuss-type diffusions fabricated on Al_2O_3 walls and light management in TiO_2 layers. Finally, hybrid electrolyte systems are proposed to fabricate high-performance DSCs containing ionic liquid-type electrolytes.

Experimental

Solar cell fabrication.—Ti-Nanoxide D paste (Solaronix SA) was coated on SnO_2/F layered glasses ($30 \Omega/\square$, Nippon Sheet Glass Co., Ltd.). The substrate was baked at 450°C to fabricate $11 \mu\text{m}$ thick TiO_2 layers. The substrate was dipped in the solution containing cis-bis(isothiocyanato)bis(2,2'-bipyridyl-4,4'-dicarboxylato)-ruthenium(II) bis-tetrabutylammonium [(Ruthenium 535 bis-TBA, N719 (Solaronix)]. The concentration was fixed to 38 mM in *t*-butylalcohol and acetonitrile [1:1 (vol/vol)]. Pt sputtered SnO_2/F layered glass substrates were employed as counter electrodes. Electrolyte solution was injected into the Al_2O_3 films (100 and 200 nm pore size, Al_2O_3 wall: pore = 1:1, volume, for both films, Whatman) at room temperature. The surface-unmodified Al_2O_3 film or surface-modified Al_2O_3 film filled with electrolytes was sandwiched between a TiO_2 anode and a counter electrode (cell 3 in Fig. 2). Cell 1 without Al_2O_3 films was also prepared as the reference (cell 1 in Fig. 2). Electrolyte (IL300) consisted of I_2 (300 mM), LiI (500 mM), *t*-butylpyridine (580 mM) in 1-methyl-3-propylimidazolium iodide (MePrImI) containing 5% water. The cell area was 0.25 cm^2 . The solar cell measurement was carried out by using a photomask on the solar cell to remove the effect of optical reflection.

To compare the effect of the straight ion path for Al_2O_3 films, random ion-path fabricated with Al_2O_3 nanoparticles (Fig. 2 Cell 2) was prepared as the reference. Ionic liquid type electrolyte IL300 and nanoparticles (100:40 wt %) were put on a mortar and were well mixed mechanically to prepare soft gel type electrolytes. This was coated on the TiO_2 electrode and a counter electrode was set on the soft gel type electrolytes (cell 2 in Fig. 2).

Surface modification.—Surface modification of Al_2O_3 films was carried out in the same method reported previously (Fig. 3). Porous Al_2O_3 film was dried at 450°C for 30 min before the reaction. After cooling to room temperature, the Al_2O_3 film was dipped into 10% solution of 12-iodododecanoic acid (C12-I, kindly donated from Manac Inc.) (35 g) in acetonitrile (100 mL) under N_2 . The mixture was kept for one night in the dark at room temperature. The film was taken out and washed with acetonitrile to remove unreacted C12-I. After that, the film was dried at room temperature under a vacuum for one night to obtain Al-C12-I film. Al-C12-I film was dipped into 10% solution of 1-methylimidazole in acetonitrile for one night at

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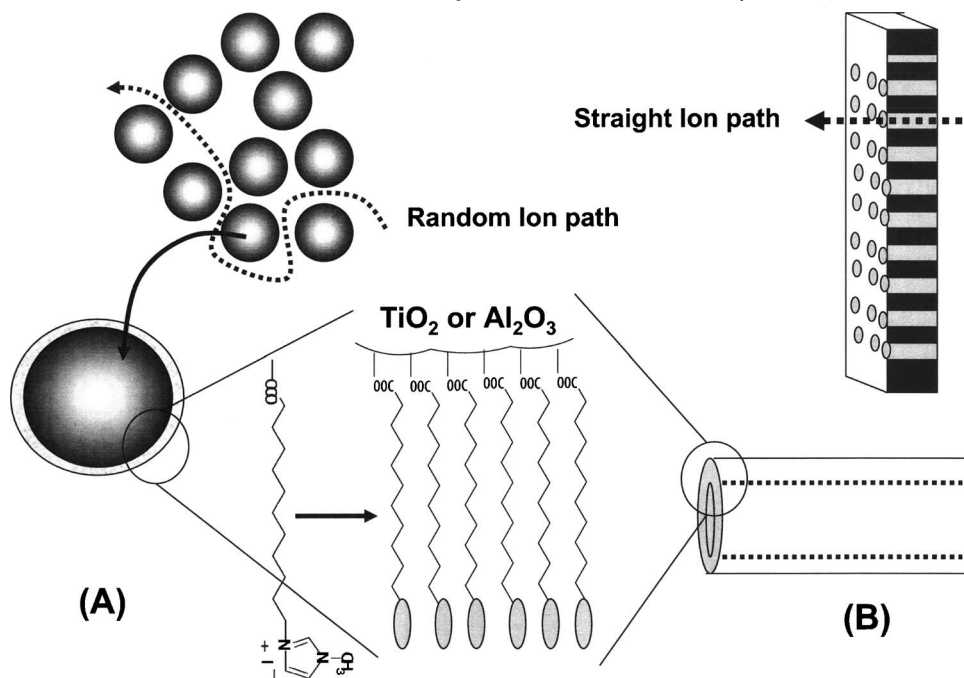


Figure 1. Schemes for random ion paths and straight ion paths. (A) Random ion paths.³⁵ (B) Straight ion paths using anodic oxidation Al_2O_3 film. Both are surface modified with imidazolium moieties bearing a long alkyl group. See Fig. 3.

room temperature under N_2 . The film was taken out and washed with acetonitrile and dried under a reduced pressure to obtain Al-C12Im-I film.

Measurement.—Photoelectrochemical measurements were performed using a solar simulator (XHP-1, Bunko-Keiki Co. Ltd., AM 1.5, 100 mW/cm^2) equipped with a xenon lamp (XLS-150A). IPCE (incident photon-to-current efficiency) was measured by using Bunko-Keiki model KHP-1 equipped with a xenon lamp (BPS-X300B).

The apparent diffusion coefficient was measured by limiting current methods. The electrode size was 0.25 cm^2 . The symmetrical cells were fabricated by sandwiching the porous Al_2O_3 film with two Pt electrodes fabricated on SnO_2/F layered glass substrates. When one Pt electrode is charged negatively, I_3^- diffuses to near the electrode and is reduced to I^- . When the potential is increased negatively, the current increases linearly and levels off. This current is called limiting current. I_3^- diffusion is slower than that of I^- because

the I_3^- is larger than I^- . Therefore, I_3^- is responsible for the limiting current. The apparent diffusion coefficient was calculated by using the formula³⁶⁻³⁸

$$D = I_{\text{lim}} d / (2nFC) \quad [1]$$

where D is the diffusion coefficient, I_{lim} is the limiting current, d is the cell gap, n is the number of electrons (2), F is the Faraday constant, and C is the initial I_2 concentration.

The amount of iodide adsorbed on surface-unmodified Al_2O_3 films or surface-modified Al_2O_3 films was measured as follows. I_2 (300 mM) was dissolved in MePrImI. Al-C12Im-I film or unmodified Al_2O_3 film was dipped in the solution for 24 h. $\text{I}_2(\text{I}_3^-)$ concentration in the supernatant solution was analyzed by a UV and visible absorption spectrometer. The presence of I_3^- of Al-C12Im-I films or the unmodified Al_2O_3 films was observed by a JASCO model NRS-5000 Raman spectrometer (Ar laser-GLS2162-514.5 nm) after these films were taken out of MePrImI solution containing I_2 (300 mM).

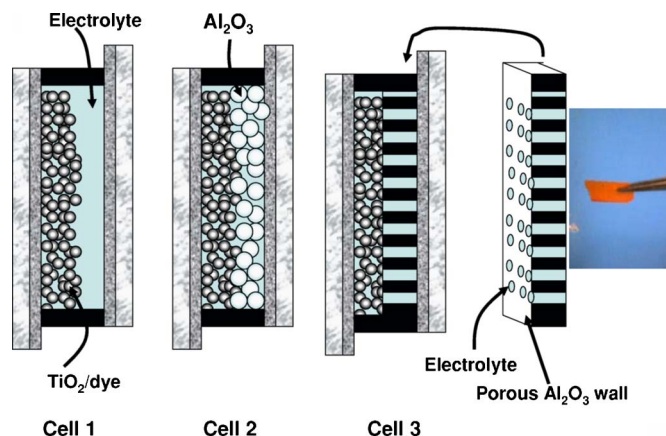


Figure 2. Abbreviations for various cell structures. Cell 1: Cell structure without Al_2O_3 film (conventional cell). Cell 2: Cell structure with Al_2O_3 particles (reference cell). Cell 3: Cell structure with surface modified or surface unmodified Al_2O_3 film. Inset: an Al_2O_3 film filled with IL300 electrolyte.

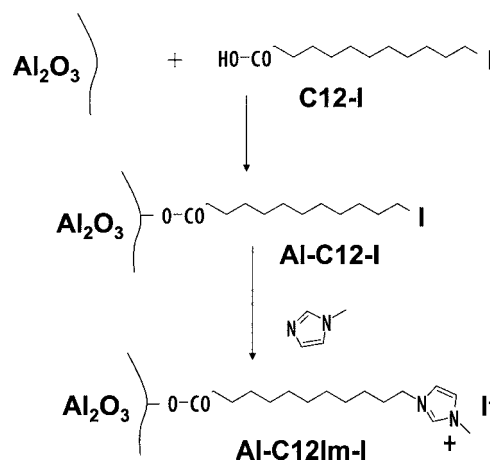


Figure 3. Synthesis of surface modified anodic oxidation Al_2O_3 film.

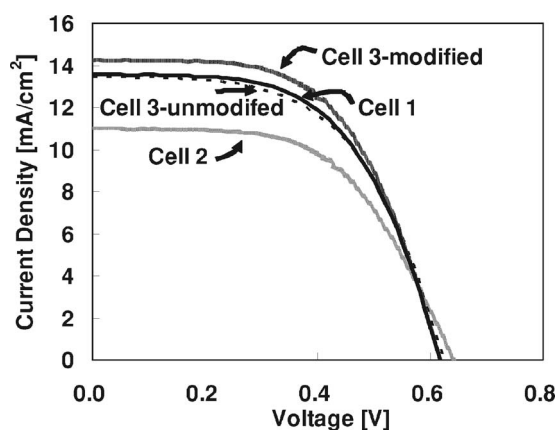


Figure 4. Relationship between photocurrent density and photovoltage for DSCs cell 1. Cell 1 filled with IL300 electrolyte Cell 2. Cell 2 filled with the mixture of Al_2O_3 nanoparticles and IL300 (40:100) (soft gel) Cell 3-unmodified. Cell 3 equipped with surface unmodified anodic oxidation films filled (diam 100 nm) with IL300 Cell 3-modified. Cell 3 equipped with surface modified anodic oxidation films (diam 100 nm) filled with IL300 5×5 mm, AM1.5, 100 mW/cm^2 , photomasked.

Results and Discussion

Comparison of straight ion path with random ion path (unmodified Al_2O_3 nanoparticles and unmodified anodic oxidation Al_2O_3 films).— The picture of a surface-modified Al_2O_3 film filled with IL300 is shown in the insert in Fig. 2. The pore size was 100 nm. The ionic liquid-type electrolyte was held in the pores and did not leak from the films.

Figure 4 shows the I - V curves for a conventional cell 1, soft gel type. Cells 2 and 3 equipped with unmodified anodic oxidation Al_2O_3 film (cell 3-unmodified). Photovoltaic performance for cell 3-unmodified was almost the same as that for cell 1. However, the photovoltaic performance of cell 2 was lower than that for cell 1. The whole hole volume (volume filled liquid electrolytes) against electrochemically inactive Al_2O_3 wall in the Al_2O_3 films is about 1:1 (volume) for cell 3-unmodified; however, the ratio is about 10:1 (volume) for cell 2. Even when the electrolyte volume ratio for cell 3-unmodified was less than that for cell 2, the cell performance was better than that for cell 2, suggesting the effectiveness of straight ion diffusions in anodic oxidation Al_2O_3 films. We have already reported that the photovoltaic performance decreased largely with an increase in the amount of nanoparticles added in the electrolytes in order to fabricate solidified DSC.³⁵ Short circuit current (Jsc) decreased to a half when 25 vol % of unmodified nanoparticle was added to ionic liquid-type electrolytes.³⁵ The fact that cell 3-unmodified containing 100 vol % of Al_2O_3 wall against electrolytes shows the same photovoltaic performance as cell 1 is remarkable.

Comparison of cell 3-unmodified and cell 3-modified.— Figure 4 also shows the photovoltaic performance for cell 3-modified. The straight nanoholes in Al_2O_3 films were surface-modified with imidazolium moieties bearing a long alkyl group (Al-C12Im-I), because the modification by using this molecule gave the best results in our previous report.³⁵ Jsc for cell 3-modified increased by about 1 mA/cm^2 from that for cell 3-unmodified. We have already reported that the surface modification of nanoparticles in cell 2 was very useful in order to retard the decrease in Jsc of the cell after solidification.³⁵ This shows that surface modification with imidazolium moieties bearing long alkyl groups is also useful for the anodic oxidation Al_2O_3 films. Figure 5 shows the apparent I_3^- diffusion coefficient of electrolyte layers for ionic liquid-type electrolyte (IL300 only) (cell A), an unmodified Al_2O_3 film filled with IL300 (cell B) and a modified Al_2O_3 film filled with IL300 (cell C). In

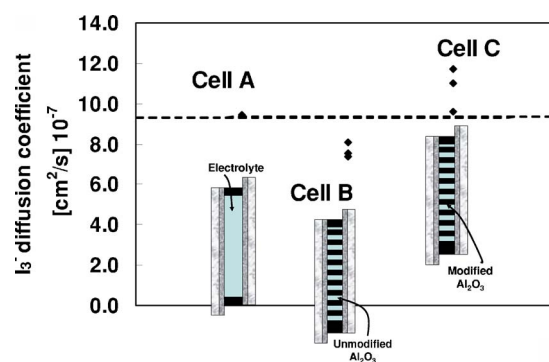


Figure 5. Apparent I_3^- diffusion coefficient in liquid electrolyte and quasi-solid electrolytes. Cell structure A: $\text{SnO}_2/\text{F}/\text{Pt}/\text{IL300}/\text{Pt}/\text{F}/\text{SnO}_2$. Cell structure B: $\text{SnO}_2/\text{F}/\text{Pt}/\text{unmodified anodic oxidation } \text{Al}_2\text{O}_3 \text{ film (200 nm) filled with IL300}$. Cell structure C: $\text{SnO}_2/\text{F}/\text{Pt}/\text{modified anodic oxidation } \text{Al}_2\text{O}_3 \text{ film (200 nm) filled with IL300}$.

order to calculate the apparent diffusion coefficient, the presence of the Al_2O_3 wall was not taken into consideration. The apparent I_3^- diffusion coefficient for unmodified Al_2O_3 films containing IL300 (cell B) was a little lower than that of IL300 only (no Al_2O_3 film) (cell A). However, modified Al_2O_3 films with IL300 (cell C) had higher apparent diffusion coefficient than IL300 only (cell A) and unmodified Al_2O_3 films filled with IL300 (cell B). The higher Jsc for cell 3-modified can be explained by the higher apparent diffusion coefficient.

Concentration of $\text{I}_2(\text{I}_3^-)$ species on Al_2O_3 surfaces.— It has been reported that $\text{I}_2(\text{I}_3^-)$ molecules are concentrated on nanoparticles in soft gels.²¹ $\text{I}_2(\text{I}_3^-)$ concentration of supernatant solution was measured after anodic oxidation Al_2O_3 films were dipped in the iodine solution as explained above. The $\text{I}_2(\text{I}_3^-)$ concentration was calculated from the $\text{I}_2(\text{I}_3^-)$ concentration of the supernatant solution. $\text{I}_2(\text{I}_3^-)$ concentration decreased from 300 to 269 mM after Al_2O_3 films were dipped. This means that 1493 mM of $\text{I}_2(\text{I}_3^-)$ is concentrated by adsorption on the surface of nanopores in the modified Al_2O_3 films. In addition, it was observed by a Raman spectrometer that I_5^- species (145 cm^{-1}) were formed on the Al_2O_3 surfaces as shown in Fig. 6. This I_5^- species were not observed in IL300 electrolyte. It has been already reported by Yanagida and his co-workers that I_5^- is also present on unmodified nanoparticles in soft gels.²¹ The presence of I_5^- also supports the condensation of I_2 species on the modified Al_2O_3 films.

These experimental results strongly suggest that $\text{I}_2(\text{I}_3^-)$ is condensed on chemically modified straight pores of Al_2O_3 films. We believe that $\text{I}_2(\text{I}_3^-)$ are concentrated on the surfaces and makes ion

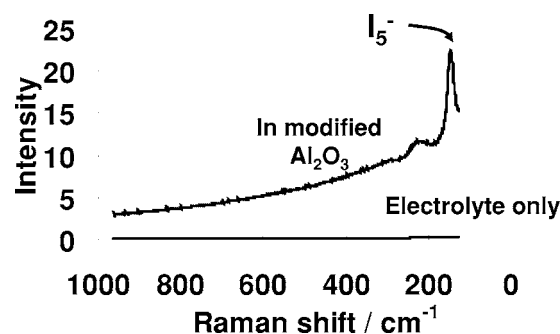


Figure 6. Raman spectra for IL300 only and anodic oxidation film filled with IL300 145 cm^{-1} corresponds to I_5^- . I_5^- (110 cm^{-1}) was not observed because of the measurement limitation of the apparatus.

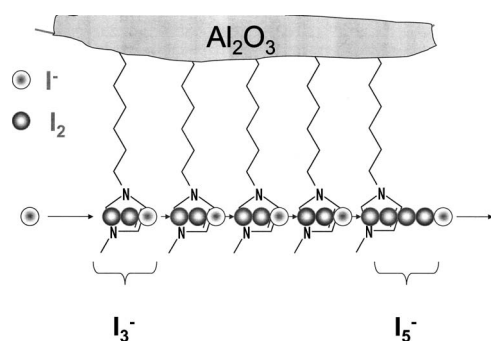


Figure 7. Schematic explanation of concentrated $I_2(I_3, I_5^-)$ along imidazolium moieties and Gröthhuss mechanism.

paths, where, $I_2(I_3^-)$ moves by Gröthhuss mechanism as shown in Fig. 7. This is why apparent diffusion coefficient was kept so high even when the IL300 was quasi-solidified and the content of electrochemically inactive Al_2O_3 wall reached 50 vol % in the electrolyte layer. If the Al_2O_3 wall content is taken into consideration, the I_3^- diffusion coefficient would increase to twice. It has been already reported that Gröthhuss mechanism type diffusion of $I_2(I_3^-)$ occur when I_2 concentration is extremely high even in liquid electrolytes. This also supports the mechanism shown in Fig. 7.

Light management in TiO_2 layers.—IPCE values for cell 1 was lower at around 400 nm than those for cell 3-modified as shown in Fig. 8. The decrease in IPCE at around 400 nm was one of crucial problems for cells filled with ionic liquid-type electrolytes, because more than 300 mM of I_2 had to be added to increase conductivity in the electrolyte layers. This amount of I_2 is about ten times higher than that in the case of organic liquid type electrolytes (30–50 mM). Because of the UV absorption of $I_2(I_3^-)$ species, light penetration depth at around 400 nm decreases and IPCE at around 400 nm decreases. The improved IPCE of the cell 3-modified at around 400 nm can be explained as follows. $I_2(I_3^-)$ is adsorbed on surface-modified Al_2O_3 films. On the other hand, the $I_2(I_3^-)$ concentration of ionic liquid-type electrolytes in TiO_2 layers decreased relatively. Because of this, light penetration depth increased in TiO_2 layers. The decrease in $I_2(I_3^-)$ concentration is expected to decrease conductivities of electrolytes in pores of TiO_2 layers. However, this may not be a crucial problem because of the thin TiO_2 layers.

Concept of hybrid electrolyte.—The role of the surface-modified Al_2O_3 films is summarized in Fig. 9. Surface-modified straight holes in Al_2O_3 layers adsorb $I_2(I_3^-)$ species and make ion

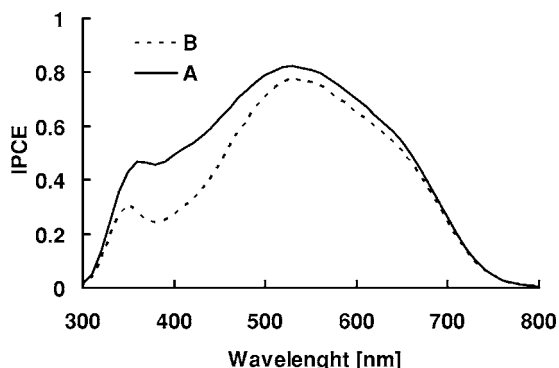


Figure 8. IPCE curves for cell 1 and cell 3-modified filled with IL300 A. Cell 3 equipped with surface-modified-anodic oxidation Al_2O_3 films filled with IL300 electrolyte B. Cell 1 filled with IL300 electrolyte.

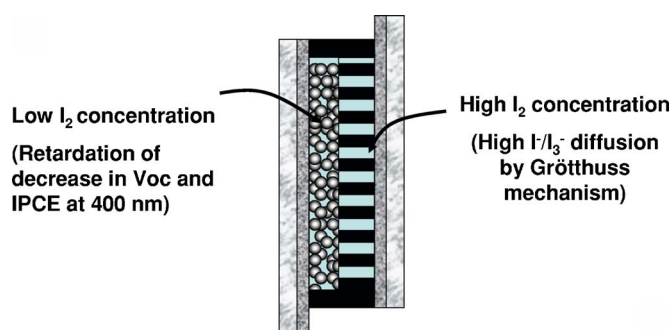


Figure 9. Concept on hybrid electrolyte containing high I_3^- concentration in Al_2O_3 layer and low I_3^- concentration in TiO_2 layer.

paths, which realizes high diffusions of $I_2(I_3^-)$ aided by Gröthhuss mechanism even when electrochemically inactive Al_2O_3 occupied almost half of the electrolytes. Because of concentration of I_2 on Al_2O_3 walls, relative concentration of $I_2(I_3^-)$ species in nanopores of TiO_2 layers decreases. This increased IPCE at around 400 nm. The I_2 -concentrated layers in Al_2O_3 films act as high conductive layers. The hybrid electrolyte system containing high $I_2(I_3^-)$ concentration in Al_2O_3 layers and low $I_2(I_3^-)$ concentration in TiO_2 layers is one of the solution in order to prepare high-efficiency DSCs filled with ionic liquid-type.

Conclusion

To solidify the ionic liquid-type electrolytes, electrolytes were held in the straight nanopores in anodic oxidation Al_2O_3 films. The quasi-solidified DSCs had higher photovoltaic performances than the parent liquid electrolyte-type DSCs even when electrochemically inactive Al_2O_3 occupied 50% in the electrolyte layer. The higher efficiency was explained by the fabrication of ion paths on modified Al_2O_3 surfaces, where $I_2(I_3^-)$ species were concentrated and were expected to diffuse by Gröthhuss mechanism. In addition, $I_2(I_3^-)$ concentration in nanopores of TiO_2 layers decreased relatively, which increased light penetration depth at around 400 nm in TiO_2 layers and increased IPCE at around 400 nm. It is concluded that the concept on composite electrolyte systems containing high $I_2(I_3^-)$ concentration in Al_2O_3 straight pores and low $I_2(I_3^-)$ concentration in TiO_2 layers has the potential to realize high-performance quasi-solidified DSCs.

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